

# 09: Intro to Multiple Linear Regression

Stat 230 – Spring 2026

## Note

This is the tutorial version of the R activity. If you prefer to use .qmd and RStudio directly, you can use <https://aluby.github.io/stat230/activities/09-mlr>

## 1 Data

- In an effort to understand the important aspects of a satisfactory supervisor, clerical employees at a large financial organization were asked to rate their immediate supervisor.
- The survey questions were designed to measure overall performance of the supervisor, as well as six additional characteristics.
- Employees were asked to rate the following statements on a scale from 0 to 100 (0 = “completely disagree”, 100 = “completely agree”)

*Note:* this data comes from the textbook

| Variable   | Description                                                                 |
|------------|-----------------------------------------------------------------------------|
| rating     | Overall rating of supervisor performance                                    |
| complaints | Score for “Your supervisor handles employee complaints appropriately.”      |
| privileges | Score for “Your supervisor allows special privileges.”                      |
| learn      | Score for “Your supervisor provides opportunities to learn new things.”     |
| raises     | Score for “Your supervisor bases raises on performance.”                    |
| critical   | Score for “Your supervisor is too critical of poor performance.”            |
| advance    | Score for “I am not satisfied with the rate I am advancing in the company?” |

**Before looking at the data**, how do you think each predictor variable will be related to rating? What do you think will have the strongest relationship? The weakest?

## 1.1 Loading data

To load the supervisor data set, run the following code chunk:

```
supervisor <- read.csv("http://aluby.github.io/stat230/data/supervisor.csv")
```

## 1.2 EDA: Making a pairs plot

We will use the `ggpairs()` function in the `{GGally}` package to make the pairs plot. By default, it will include every variable in the dataset. The code below firsts `select()`s the columns we want to include (all of them). I have also moved the response variable to be the last in the list, so that it shows up on the Y axis.

```
supervisor |>
  select(complaints, privileges, learn, raises, critical, advance, overall) |>
  ggpairs()
```

- Which predictor variables appear to be most closely related to `overall`?
- Which predictor variables appear to be most closely related to *each other*?

## 1.3 SLR

In the code chunk below, choose what you think will be the best *single* predictor and fit an SLR model. Interpret the intercept and slope coefficient and report the  $R^2$  value.

```
supervisor_slr <- lm(overall ~ _____, data = supervisor)
summary(supervisor_slr)
```

## 1.4 MLR

In the code chunk below, use all predictors and fit an MLR model. Interpret the intercept and two of the slope coefficients and report the  $R^2$  value.

```
supervisor_mlr <- lm(overall ~ _____, data = supervisor)
summary(supervisor_mlr)
```

## 2 Predicting responses

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We can use most of our SLR tools for MLR. For example, here's some code I could use to calculate predictions. (Uses the bikes data)

```
predict(bikes_lm, newdata = data.frame(temp_actual = 90, windspeed = 6))
```

- Use this as starter code to predict the overall manager score for a manager with scores of 50% for all predictors
- What if the manager from the previous question decides to try to improve her critical score by 10 points (so 40% instead of 50%) and her complaints score by 10 points (so 60% instead of 50%). How does that change the prediction?

```
# your code here
```

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### 2.1 Function quick reference

| Function                                         | Purpose                                  |
|--------------------------------------------------|------------------------------------------|
| lm(formula, data)                                | Fit a MLR model                          |
| predict(model, newdata, interval, level, se.fit) | Make predictions and calculate intervals |
| data  > select(vars)  > ggpairs()                | Make pairs plot for selected variables   |